

EXERCISE 1

Problem 1 (Sum of Dice). When rolling two six-sided dice, what is the probability that the sum of the dice is greater than or equal to 9?

Problem 2 (Product of Dice). When rolling three six-sided dice, what is the probability that the product of the dice is *not* a prime number?

Problem 3 (Wrong Desserts). Alice, Bob, and Clara go to a restaurant. For dessert, Alice orders an apple pie, Bob chooses brownies, and Clara orders a cheesecake. When the waiter brings the desserts, he places them randomly in front of the three friends. What is the probability that none of the three gets their ordered dessert?

Problem 4 (Distinct Digit Pair Time). You have an analog watch, but you forgot to wind it, causing the watch to stop. The time is recorded in the format HH:MM (hours and minutes), where HH ranges from 00 to 11 and MM ranges from 00 to 59.

What is the probability that the displayed time consists of strictly increasing digits?

For example, the times 01:23 and 04:57 are of that form, but 04:25, 11:11, and 03:34 are not.

Problem 5 (Coin Tosses in R). You toss a fair coin 9 times in a row and write down the sequence of heads (H) and tails (T).

What is the probability of having exactly three head and tail sequences?

For example $HHHTTTHHHH$ consists of 3 sequences, namely $3 \times H$, followed by $2 \times T$, then $4 \times H$.

Solve this problem with brute force using R.